

StreamLens

A second pair of eyes for citizen stream assessments — the citizen always decides.

Primary Track 1: Citizen Science UX · also 3 (AI-supported assessment), 2 (data to insight), 6 (early warning), 5 (community), 7 (FHIR)

LIVE DEMO

streamlens-ten.vercel.app

CODE

github.com/Mitalimehta02/streamlens

DEMO VIDEO

youtube.com/watch?v=ZC17Y4XpUPE

The API runs on a free Render instance. A scheduled ping every ten minutes keeps it awake; if it has been idle anyway, the first request takes about **42.7 s** and every request after it is under a second. No account, no sign-in, nothing to install.

THE PROBLEM

Urban streams are where ecosystem health and human health meet, and they are in trouble: across 100 urban stream sites in five European cities, the OneAquaHealth policy brief reports that **“pharmaceuticals were detected in 91% of monitored sites”**.

Monitoring that closely is expensive, so the gaps between official surveys are filled — if at all — by volunteers, who are asked ecology questions in vocabulary they were never taught.

StreamLens does not replace the expert. It makes the volunteer’s twenty minutes on the bank produce a record a scientist can actually use, and it refuses to guess on their behalf.

THE LOOP

1 · Observe

Photograph the reach upstream and downstream. A vision model drafts answers to the OneAquaHealth question set as suggestion chips, each with a confidence and a one-sentence reason pointing at something in the picture. The citizen accepts, changes or rejects every one. Plain-language glossary, six languages, works offline.

2 · Understand & act

A stream health card built only from citizen-confirmed answers. A 48-hour alert that shows its own arithmetic: the measured value, the threshold it crossed, and where that threshold came from. Observed problems map to restoration measures from the OneAquaHealth D2.4 catalogue, cited to the page.

3 · Return

Quests computed from real gaps in the record — a site nobody has seen this season, a reach that needs a second opinion. Points for evidence quality and for filling gaps, never for volume. Teams on the leaderboard, never individuals. A wellbeing mirror reported only in groups of ten or more.

JUDGE QUICK TOUR — ABOUT FOUR MINUTES

1. **Pick a site.** All 106 OneAquaHealth / ENORA research sites, across Benevento, Coimbra, Ghent, Oslo and Toulouse. No location permission needed.
2. **Choose whether the AI helps.** With it, a shrunk and EXIF-stripped copy of your photographs goes to Google Gemini under its free-tier terms. Without it, **no request is made at all** — verified in a browser, not merely promised. Both paths ask the same questions and record the same data.
3. **Photograph upstream and downstream**, or open **“Try with sample photos”** for six Creative Commons images. One has no water in it: the assessment should refuse that one rather than guess.
4. **Confirm every answer.** Where you disagree with the AI, both answers are kept. The Good / Moderate / Poor rating is never suggested.
5. **Read what came back.** *Explore* holds the health card, the alerts with their thresholds and sources, and the measures. *Community* holds coverage, teams, the wellbeing mirror and the **FHIR R4 export**.

BUILT IN THE OPEN

Monorepo: FastAPI + SQLAlchemy on Render, React + Vite PWA on Vercel, Postgres on Neon — every tier free. 279 API tests and 22 web tests, none touching the network. Lighthouse mobile on the live site: **94 performance, 100 accessibility, 100 best practices, 100 SEO**. Built 23–27 September 2026.

RESPONSIBLE AI — THE GUARD IS IN THE CODE, NOT THE PROMPT

1 · The prompt The model is told to answer “not sure” rather than guess, and to stay silent on any image with no watercourse in it.	2 · The validation gate Every code the model returns is checked against the published question set before a citizen sees it. An invented code is refused at confidence 1.0.	3 · The interface Every suggestion starts unanswered. The record stores the citizen’s answer <i>and</i> what the AI suggested — never the AI’s answer alone.	
13 / 13 images with no watercourse where it stayed silent	90% / 92% agreement, exact match and set overlap	0 invented answer codes across 187 suggestions	6 of 24 questions the AI is never allowed to suggest

Read “agreement”, not “accuracy”. There were no human expert labels. The labeller is one AI (Claude) and the model under test is another (Gemini); two systems agreeing is not either being right. The sample is 29 freely-licensed photographs and 58 scored comparisons — far too small for a confident claim in either direction. The README names the three disagreements an ecologist would catch first, including a question we reworded on a guess at intent, and asks to be corrected.

STANDARDS

Every assessment exports as **FHIR R4** — a LocationOah, one ObservationIndicatorsOah per answered question, and a Provenance naming the citizen by pseudonymous identifier and the AI as an assembler Device with its model and prompt version. Validated with the official HL7 validator against the **hl7-eu/oah** profiles: **0 errors, 0 warnings**. A negative control — one Observation deliberately broken — is committed alongside, to prove the profiles were really applied and not silently skipped.

Show all questions

CHANNEL, BED AND BANKS

Channel bed

What the bottom of the stream is made of. A natural **bed** is gravel, sand, stones or mud that the stream itself shaped. An artificial **bed** is concrete or another hard surface people laid down.

bed

Possible sewer overflow after heavy rain

28 mm of rain is forecast here over the next 48 hours, and 1 visitor report(s) in the last 14 days mentioned sewage or a discharging pipe at this site. Where sewers carry rainwater as well as foul water, heavy rain can cause them to overflow into the stream. Many authorities advise avoiding contact with affected water for around 72 hours after such an event. Check official local advice.

▼ Why this fired — the exact numbers

Where a sewer carries both foul water and rainwater, heavy rain can

The AI suggests. Each chip carries a confidence and a reason pointing at something in the photograph. Nothing is recorded until the citizen presses Correct, Not right, or Change answer.

PATH TO PRODUCTION

- **Expert in the loop.** A review queue for flagged observations — low confidence, a disagreement with a nearby visit, or an overridden refusal.
- **Identity from the OneAquaHealth app,** not a second account system, which also brings the moderation a public dataset needs.
- **Integration by data.** Answer codes and the FHIR export already target the consortium’s own definitions.
- **Cost.** \$0.0022 per photograph measured; about \$4 per 1,000 assessments a month and \$40 per 10,000.
- **Ownership.** Hand it to the citizen-science work package under the MIT licence it already carries.

Every alert shows its arithmetic. The measured value, the threshold it crossed, and where that threshold came from — including, where honest, “project judgement”.

KNOWN LIMITS

- No expert validated the evaluation; both labeller and model are AI.
- Most demo data is synthetic and every synthetic record says so. Two sites carry a planted forecast, badged **DEMO FORECAST**, so the rain rules can be shown on a dry day.
- All six languages are machine-translated and not yet reviewed by native speakers.
- No authentication: anyone who can reach the API can post an assessment.
- Photographs are not stored on the hosted demo — measured, then discarded.
- The PWA install has not been tried on a physical phone.

CREDITS

Research sites: OneAquaHealth / ENORA API, 106 sites across five cities. **Restoration measures:** D2.4 Catalogue of measures for urban aquatic ecosystems rehabilitation, Dias, M., Serra, S.R.Q. & Feio, M.J. (University of Coimbra), OneAquaHealth, Horizon Europe Grant Agreement 101086521, DOI 10.5281/zenodo.20040211, CC-BY-4.0. **Problem statement:** OneAquaHealth Policy Brief — Urban stream ecosystem health as a One Health priority, P da Silva, J.; Serra, S.R.Q.; Calapez, A.R.; Schmeller, D.S.; Freitas, Á.; Dantas, C.; Forio, M.A.E.; op den Akker, H.; Havrevold Henni, S.; Feio, M.J., 2026, DOI 10.5281/zenodo.21476580, CC-BY-4.0. **Weather:** Open-Meteo, CC-BY-4.0. **Map tiles:** © OpenStreetMap contributors, ODbL. **Sample photographs:** six Creative Commons or public domain images from Wikimedia Commons, each credited individually in the repository. **StreamLens code:** MIT.